DAVID LEE

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EDUCATION

North China University of Technology, Beijing *September 2020*
B.E. in Information Security (junior year)
Overall GPA: 3.86527 / 4.00000
Overall Grade: 90.0837 / 100.0000
Core Professional Courses: Applied Cryptography, Data Security, Image Processing, Principles of Computer Composition, Operating System, Reverse Engineering, Digital Forensic, Software Security, Network Attack and Defense

PROFESSIONAL SKILLS

Programming Languages	C/C++(3 years+, expert), Python(3 years+, expert), Rust(1 year+, experienced), Java(1 year+, intermediate), Assembly(1 year+, experienced), Matlab(1 year+, intermediate), Shell(3 years+, experienced), Markdown(5 years+, expert), LaTeX(2 years+, intermediate)
Operation Systems	GNU/Linux(CLI&GUI, 4 years+, expert), Microsoft Windows(14 years+, expert)
Languages	Native Mandarin, Fluent Professional English(CET-4 / CET-6 / IELTS 6.5)
Frameworks	OpenSSL, Django, PyQt5, PyTorch, Tensorflow, MFC, Qt
Tools	GCC, GDB, Vim, Tmux, CMake, Git, Jupyter, etc.
Others	Applied Cryptography, Machine Learning, Basic Deep Learning, Image Processing, Reverse Engineering, Software Security, Network Programming, Database, RegEx

RESEARCH AND WORK EXPERIENCES

High-Tech Innovation Laboratory of North China University of Technology [about] October 2022 - Present
Research intern *Beijing, China*

- Mentors: Prof. Dr. Lijun Wang, A.Prof. Dr. Zhengping Li, Dr. Ying Li
- Optical camera-based 3D reconstruction of spatial objects using Alicevision framework
- Completed all the works from zero to metahuman generation
- Working on research and development of preprocess algorithm for 3D reconstruction dataset

Academic Affairs Office of North China University of Technology October 2022 - December 2022
Full stack develop *Beijing, China*

- Mentor: A.Prof. Dr. Chao Wang
- Statistics and data analysis of English degree score for Master's students
- Python3 / PyQt5

Guangzhou Chenchuang Technology Development Co.,Ltd January 2022 - March 2022
Web development intern *Wuhan, China*

- WebAssembly-based streaming video decoding and object detection in real time
- WebAssembly / FFmpeg / OpenCV
- Javascript / C++ / Rust

PROJECTS AND OPEN-SOURCE EXPERIENCES

Non-intrusive Identity Authentication System Based on Gait Recognition May 2023 - August 2023 (Projected)
Team Leader / Software Architect / Backend Developer / ML Specialist

- As Team Leader: (i) Determine team division of labor based on team members' abilities.
- As Software Architect: (i) Design overall system flow. (ii) Perform modular design, including network, crypto, video capture, fake video detection, authentication.
- As Backend Developer: (i) Train and deploy deep learning models in server end. (ii) Implement network module, including protocol design and server-end **network programming**. (iii) Implement crypto module, including **secure key distribution** on public network channel and **encryption/decryption**.
- Involved techs: `Python3`, `PyTorch`, `MySQL`, `Socket`, `ByteTrack` (Object Detection), `PaddleSeg` (Silhouette Generation), `OpenGait` (Gait Extraction), LaTeX(Technical Report).
- Details can be found in system architecture & flowchart and in demo video.

HONORS AND AWARDS

- (2023.06, Regional) Third Prize in CISCN AWDP (Attack with Defense Plus) Contest
- (2023.04, National) First Prize in The 13th MathorCup Collegiate Mathematical Modeling Challenge 2023 [certificate][thesis]
- (2023.04, Provincial) First Prize in "Lanqiao Cup" National Software and Information Technology Professional Talent Competition [certificate]
- (2022.06, Regional) Third Prize in CISCN CTF (Capture the Flag) Contest
- (2022.06, National) Winning Prize in CISCN Security Project Contest
- (2022.04, Provincial) Bronze Award in the "Youth Innovation Beijing" 2022 Challenge Cup Student Entrepreneurial Plan Competition
- (2022.01, National) Position Competency Assessment Certificate of Industry and Information Technology Talents (Industrial Internet Platform Development Engineer - Intermediate - Ability Level 4) [certificate]
- (2020.06, National) Third prize in "Lanqiao Cup" Software Development and Algorithm Contest [certificate]
- (2020.04, Provincial) First Prize in "Lanqiao Cup" Software Development and Algorithm Contest [certificate]
- Scholarship for study for the academic year 21-22
- Scholarship for Practice and Innovation for the academic year 21-22
- Scholarship for Sport for the academic year 21-22
- Scholarship for Moral Character in the academic year 21-22
- Scholarship for outstanding students in the academic year 21-22
- Scholarship for study for the academic year 20-21
- Scholarship for Practice and Innovation for the academic year 20-21

LECTURES AND TALKS

- (2023.03) "Hyper-Realistic MetaHuman - The Bridge to the Metaverse in Web 3.0" for Graduate Students in School of Information [slides]
- (2022.12) "C Programming Language" for Students in School of Information [video][slides][codes]
- (2021.12) "C Programming Language" for Students in School of Information [slides][src]
- (2020.12) "C Programming Language" for Information Security Students [handouts][src]

STRENGTHS AND SELF-EVALUATION

- **Security Enthusiasts**: I have strong will to solve practical information/network security problems by using the powerful AI techniques, especially in crossing fields.
- **Fast Learner**: I can quickly use existing knowledge to explain the unknown and apply relevant competencies to unknown areas.

- **Tools Expert**: I can skillfully use various tools and complete various development tasks, good at making computers work for people.
- **Teamwork Leader**: I am capable of efficient teamwork and reasonable task allocation, maximize team efficiency, just like parallel computing systems.
- **The Innovator**: I have strong innovation ability, can come up with some novel and feasible ideas, and can be flexibly applied to related fields.
- **The Unyielder**: High work efficiency, can bear hardships, good health.

INTERESTS AND OTHER SKILLS

- Mountain Bike Enduro (6 years+, expert): See list of top 10 world ranking stages in Strava and POV videos in Bilibili,
- Hiking (2 years+, amateur)
- Road Running (2 years+, amateur): See my Strava Profile
- Trail Running (1 year+, beginner): See my Strava Profile
- Photography (8 years+, expert)
- Video Editing (8 years+, expert)

Last edit on June 20, 2023.